HSR-ACC-69 Access Node BGP Route Churn Flap Resolution SOP (Arista CLI)

Node: HSR-ACC-69 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6315

Description

This Standard Operating Procedure (SOP) addresses a Routing Protocol Flap failure scenario on the Access Node HSR-ACC-69, situated at HSR Layout Access, utilizing Arista equipment. The failure domain revolves around BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. The underlying cause of this issue can be attributed to the complex interplay between the Routing Information Base (RIB) and Forwarding Information Base (FIB) on the Arista Access Switch. As BGP routes are constantly being updated, the RIB is responsible for maintaining a consistent view of the network topology. However, when faced with a high rate of route churn, the RIB may struggle to keep pace, leading to inconsistent route selections and eventual adjacency loss in OSPF and flooding of ISIS LSPs. Furthermore, the high-speed nature of modern networking hardware, such as the Arista 7280R, can exacerbate this issue by propagating route updates at an alarming rate, overwhelming the RIB and FIB. Therefore, it is crucial to address this issue promptly to prevent network instability and potential service disruptions.

Pre-requisite Checks

- Verify the site location at HSR Layout Access is accessible to remote hands
- Check redundant link status to ensure no single point of failure

Potential Root Causes

Cause	Probability
High RIB/FIB convergence time	High
BGP route churn exceeding RIB capacity	Medium
OSPF adjacency loss due to inconsistent route selections	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP router identifier 10.10.10.1, local AS number 65000, Description: Verify BGP neighbor counts and route churn rates

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: Check OSPF adjacency status and dead timers

Command: show isis database, Expected Output: Sequence Number: 0x12345678, Description: Inspect ISIS LSP flooding patterns
```

Resolution Steps

Step 1: Step: Reset BGP neighbor connections, Command: clear bgp * soft, Description: Clear BGP tables and reset neighbor connections to force a reconvergence

Step 2: Step: Adjust OSPF adjacency timers, Command: ospf dead-interval 60, Description: Modify OSPF dead timers to reduce adjacency loss due to inconsistent route selections

Step 3: Step: Increase RIB/FIB capacity, Command: hardware capacity rib 1000, Description: Augment RIB/FIB capacity to handle high BGP route churn rates

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP router identifier 10.10.10.1, local AS number 65000, Description: Verify BGP neighbor counts and route churn rates after resolution

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: Check OSPF adjacency status and dead timers after resolution

Command: show isis database, Expected Output: Sequence Number: 0x12345678, Description: Inspect ISIS LSP flooding patterns after resolution
```

Roll-back Procedures

- Step: Restore original BGP configuration, Command: copy running-config startup-config, Description: Restore the original BGP configuration after resolution

- Step: Reset OSPF adjacency timers, Command: no ospf dead-interval 60, Description: Reset OSPF adjacency timers to their original values after resolution

- Step: Decrease RIB/FIB capacity, Command: hardware capacity rib 500, Description: Decrease RIB/FIB capacity to its original value after resolution